

TECHNICAL PRESENTATION

Steganography vs Cryptography: Hidden Security in the Digital World

Cryptography and Network Security (CSA51)

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What is Information Security?

Information security is the practice of protecting information from unauthorized access, modification, disclosure, and destruction.

Two important techniques are:

- Cryptography
- Steganography

Both protect sensitive information but use different approaches.

Image idea:

- Lock (Cryptography)
- Hidden Image (Steganography)

What is Cryptography?

Definition

Cryptography converts readable data (plaintext) into unreadable data (ciphertext) using an encryption algorithm and a key.

Process

Plain Text



Encryption



Cipher Text



Transmission



Decryption



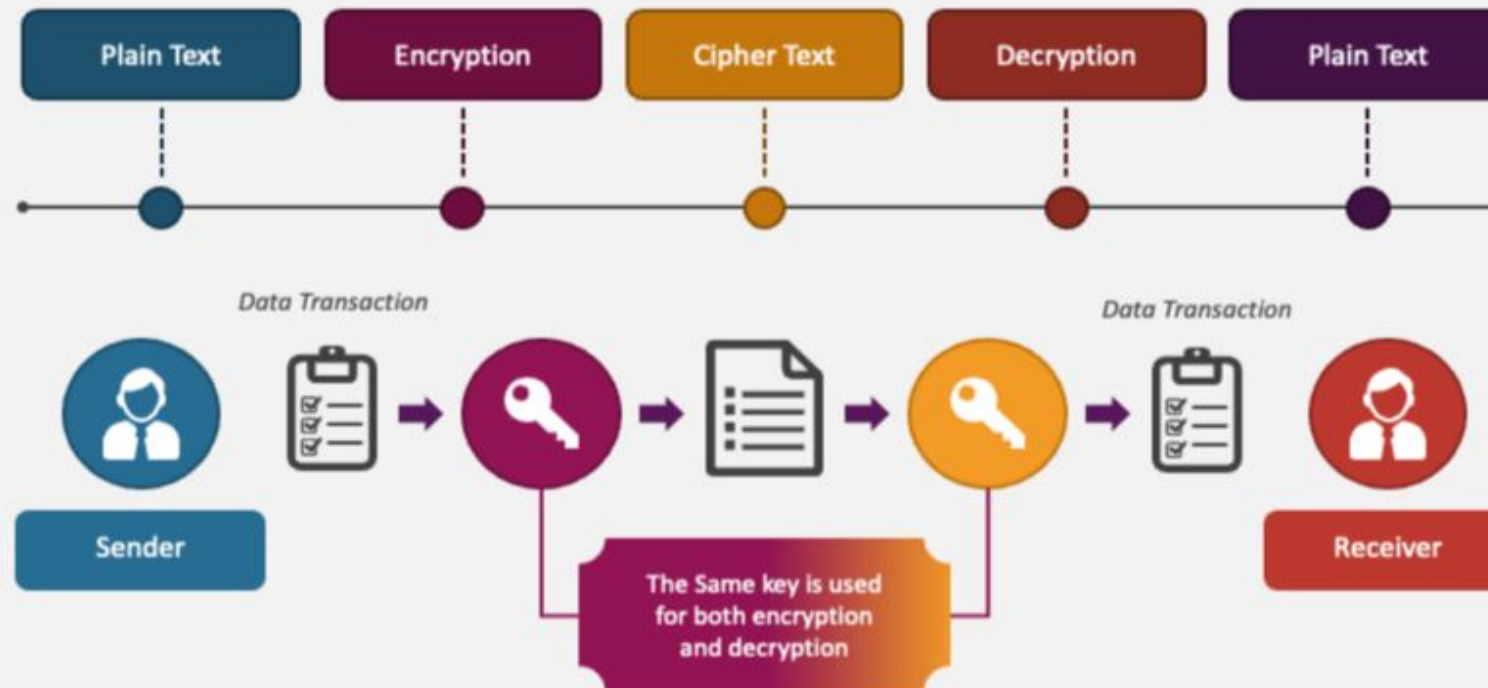
Original Message

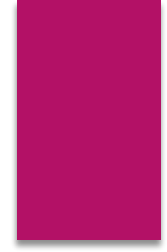
Advantages

- Confidentiality
- Authentication
- Integrity
- Non-repudiation

CRYPTOGRAPHY

Symmetric-Key Cryptography





What is Steganography?

Definition

Steganography is the art of hiding secret information inside another digital object so that no one knows a secret message exists.

Carrier can be

- Image
- Audio
- Video
- Text

Process

Secret Message



Hide inside Image



Stego Image



Receive Image



Extract Secret Message

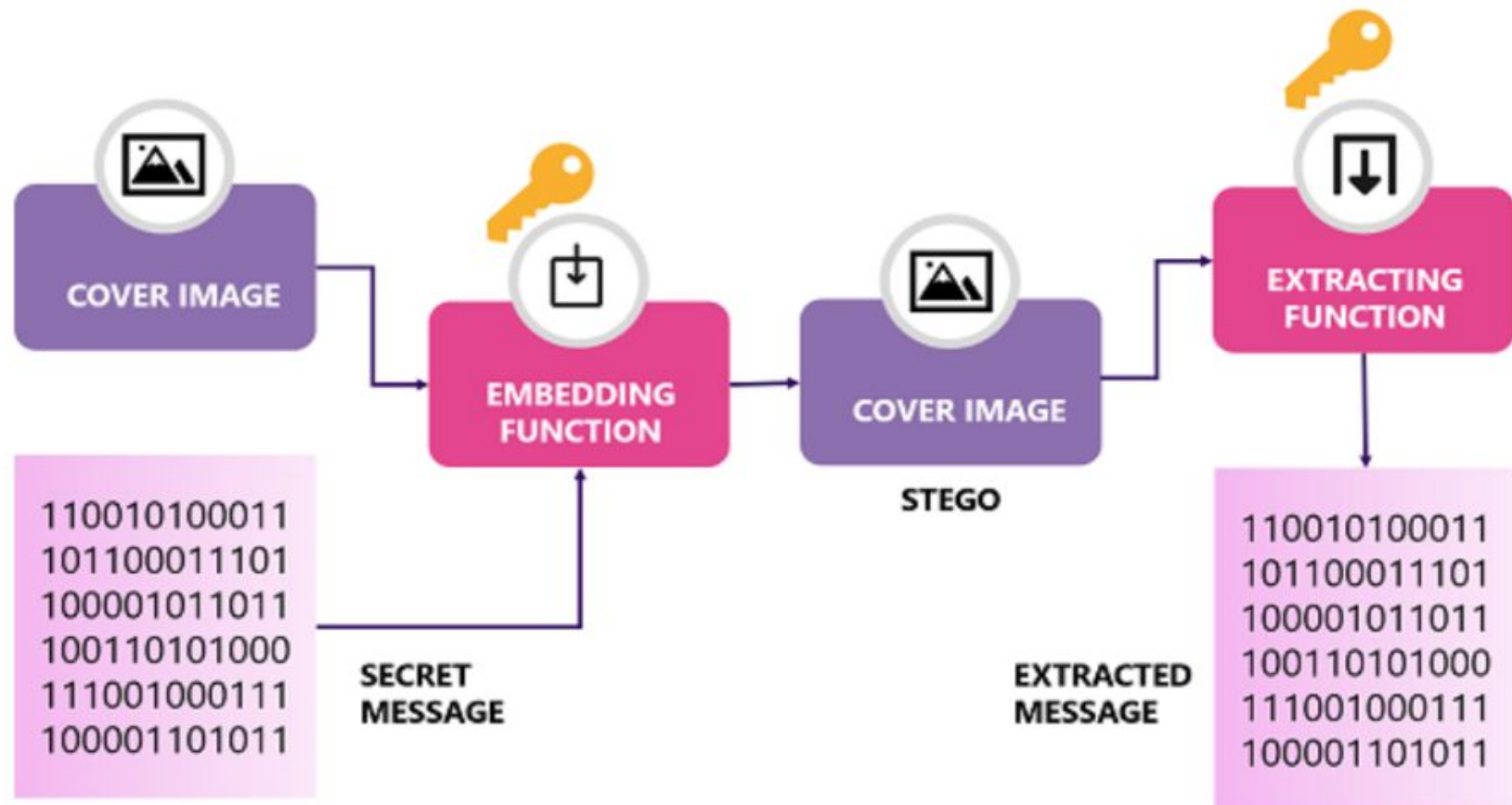


Fig 1: Functional diagram of a steganography system which works on images

Cryptography vs Steganography

Cryptography	Steganography
Encrypts data	Hides data
Everyone knows encrypted data exists	Nobody knows secret data exists
Produces ciphertext	Produces normal-looking image
Focuses on confidentiality	Focuses on secrecy
Easy to detect encrypted data	Difficult to detect hidden data

Combining Both Techniques

Best security is achieved by combining both.

Message



Encrypt



Cipher Text



Hide inside Image



Stego Image



Extract

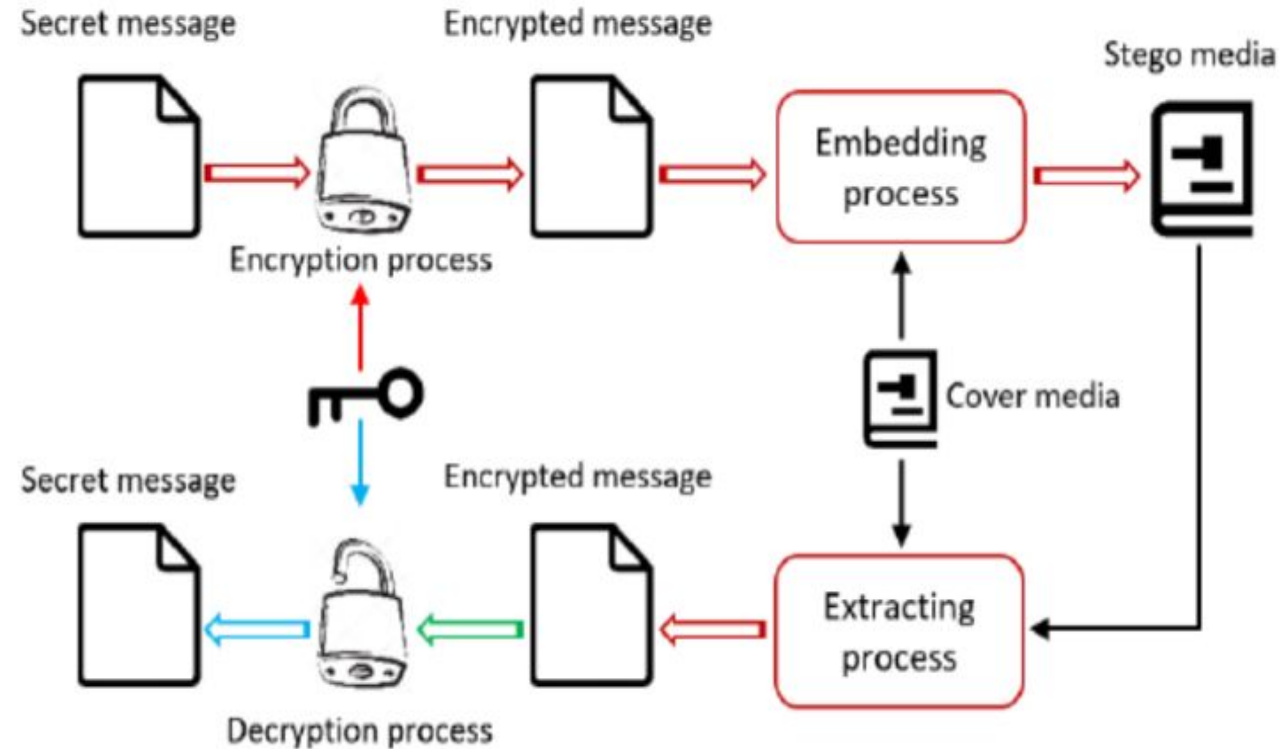


Decrypt

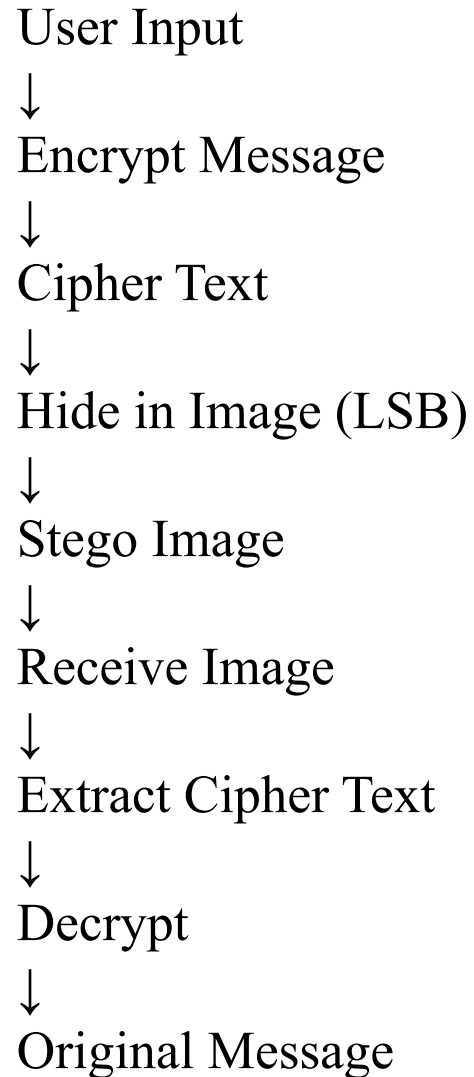


Original Message

This provides two layers of protection.



Proposed System Flowchart



Experimental Setup

Hardware

- Laptop
- 8 GB RAM

Software

- Python 3.12
- VS Code
- OpenCV
- Pillow
- NumPy

Algorithm

- Caesar Cipher (or Vigenère Cipher)
- LSB Image Steganography

Applications

- Military Communication
- Banking
- Medical Records
- Digital Watermarking
- Cybersecurity
- Cloud Storage
- Intelligence Agencies

Advantages and Limitations

Cryptography

Advantages

- Strong encryption
- Authentication
- Integrity

Limitations

- Ciphertext attracts attackers
- Key management required

Steganography

Advantages

- Hidden communication
- Difficult to detect
- Extra secrecy

Limitations

- Limited storage capacity
- Can be destroyed by image compression

Conclusion

- Cryptography protects message content through encryption.
- Steganography conceals the existence of the message.
- Combining both techniques provides stronger security than using either alone.
- The practical demonstration shows secure communication with two layers of protection.
- These methods are widely used in modern cybersecurity applications.

References

- William Stallings – *Cryptography and Network Security*
- Behrouz A. Forouzan – *Cryptography and Network Security*
- NIST Cryptographic Standards
- IEEE Xplore Digital Library
- Springer Research Articles